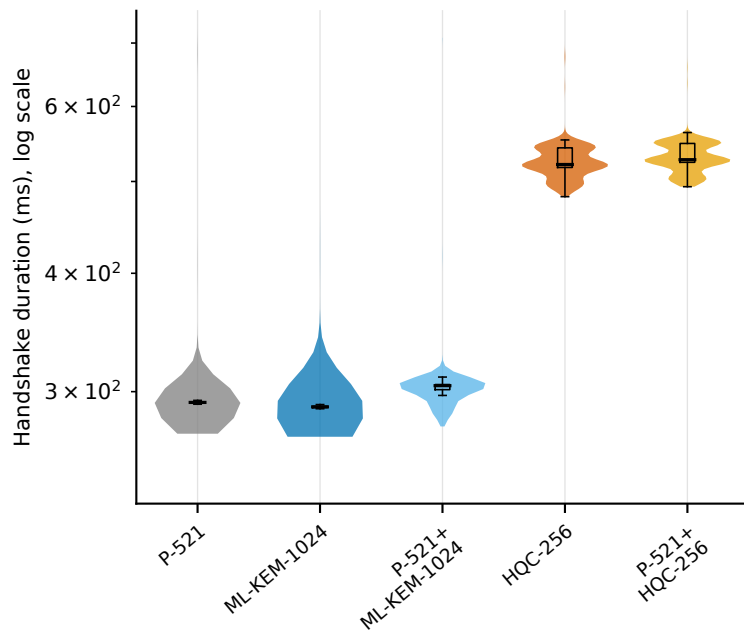
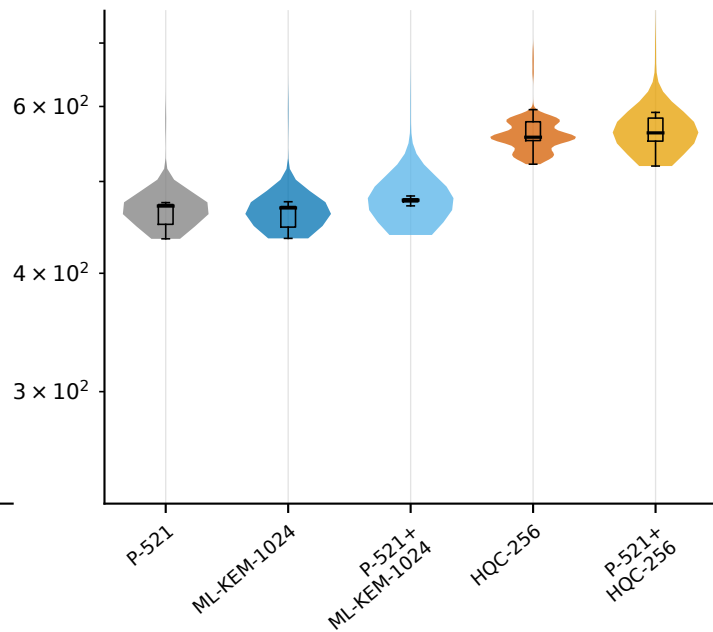


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	294.0	17.9	1.31k	469.0	12.8	1.77k
ML-KEM-1024	291.4	26.0	1.48k	467.0	13.8	1.76k
P-521+ML-KEM-1024	302.2	6.7	709.0	483.4	20.8	1.79k
HQC-256	525.3	5.0	784.4	560.5	4.5	816.7
P-521+HQC-256	531.0	4.2	798.4	570.8	15.9	1.98k